

Proofs for $R(s, t, H)$ where H is a purple subgraph

Nozomi Giladi

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1 Defining Terms:

A purple edge is an edge which can form a red or a blue subgraph. Generally, adding purple edges leads to higher chances of red and blue subgraphs forming, causing Ramsey numbers to generally be lower. Previous works have found that one purple edge is sufficient to reduce the Ramsey number $R(3, 3)$ to 5. Throughout this document, the notation $R(s, t, H)$, denotes the lowest integer, such that any colouring of K_n containing the purple subgraph H has either a blue/purple copy of K_s or a red/purple copy of K_t . The notation $R(s, H)$ will be used interchangeably with $R(s, s, H)$.

2 Proving $R(3, C_n) = R(3, n, C_n) = n$ if $3 \leq n \leq 5$:

Proof: Let there be a complete graph with the vertices $v_1, v_2, \dots, v_{n-1}, v_n$, where $n \geq 3$. It is trivial that a purple C_3 guarantees both a red and blue triangle. We will define our Hamilton cycle as containing the purple edges, $v_1v_2, v_2v_3, \dots, v_{n-1}v_n, v_nv_1$ up to swapping of vertex connections. Notably, to avoid a blue K_3 , every edge between two purple edges must be coloured red. For an example, the edge v_2v_3 must be red, because a blue K_3 would otherwise be formed from the purple edges v_1v_2 and v_2v_3 . This leads to every remaining edge being coloured red. This leads to a red K_n . It is worth noting that when n is higher than 5, it can no longer be assumed that every edge forms a triangle with two purple edges.

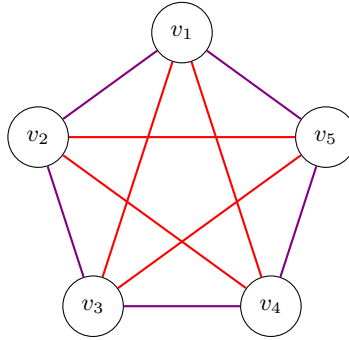


Figure 1: The K_5 counterexample with the colouring described in the proof

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3 Proving that $R(4, C_4)$ only contains one structurally unique counterexample (up to swapping of colours and relabeling of vertices):

Proof: Suppose there is a complete graph with the vertices v_1, v_2, v_3, v_4 , with the purple Hamiltonian cycle $v_1v_2, v_2v_3, v_3v_4, v_4v_1$. It is worth noting that a relabeling of vertices would not change

the core argument. Suppose that v_1v_3 and v_2v_4 are the same colour. This would form a monochromatic K_4 . Therefore, by the pigeonhole principle, it can be assumed that there is one blue and one red edge. Further arguments using this K_4 as a subgraph can be made up to swapping, and would halve the number of required computations.

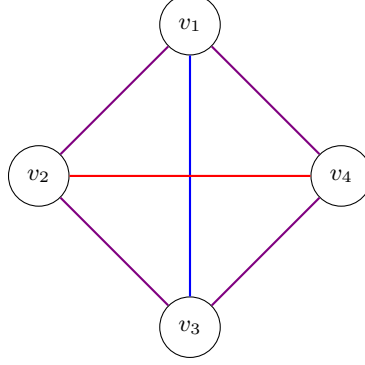


Figure 2: The K_4 counterexample with the colouring described in the proof

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4 Proving that $R(4, C_4) > 5$:

Firstly, it is a necessity to prove that $R(4, C_4) > 5$. A counterexample for K_5 can be found below. To find $R(4, C_4)$, it is essential to understand how counterexamples for K_5 were constructed.

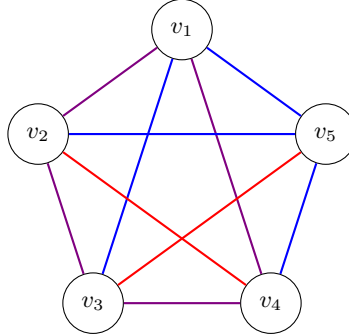


Figure 3: A K_5 counterexample. Please note this is not the only counterexample and non-isomorphic counterexamples do exist.

Firstly, a new vertex v_5 was added. Once again, we construct sets of edges which cannot all be a certain colour. At least one element of the set $\{v_1v_5, v_3v_5, v_4v_5\}$ must be blue, since we would otherwise have a blue K_4 . Similarly, at least one element of the set $\{v_2v_5, v_3v_5, v_4v_5\}$ must be blue. The same principle can be applied to find that there is at least one red edge in the sets $\{v_1v_5, v_2v_5, v_3v_5\}$ and $\{v_2v_5, v_3v_5, v_4v_5\}$. It then logically follows that there is at least one red edge and one blue edge in the set. $\{v_1v_5, v_2v_5, v_3v_5, v_4v_5\}$.

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5 Proving that $R(4, C_5)$ only contains one structurally unique counterexample:

Proof: We are trying to find a positive integer n such that any K_n with a purple C_5 subgraph would contain a blue K_4 or red K_4 . This result will be verified computationally with a Julia program, but

the uniqueness of the counterexample will be proved here. If we assume one counterexample for K_5 (up to relabeling and colour swapping), we can reduce the necessary computations by a factor of 32.

We will first define the smallest possible value for n where this graph could exist. The smallest graph where a purple C_5 could exist is K_5 . Let there be a K_5 with the vertices v_1, v_2, v_3, v_4 , and v_5 . We will now define a purple Hamilton 5-cycle with the edges

$$v_1v_2, v_2v_3, v_3v_4, v_4v_5, v_5v_1$$

up to relabeling vertices. We can now apply the inclusion-exclusion principle. It is worth noting that any non-purple subset of the form with at least two adjacent purple edges can be analysed with the following method, and generality is therefore not lost when choosing edges. Suppose that v_1v_3, v_1v_4 and v_2v_4 are the same colour. This would lead to a monochromatic K_4 . Therefore, by the pigeonhole principle, we know that two of these edges must be the same colour. Without loss of generality, (up to swapping and vertex relabeling) we will assume that v_1v_4 and v_1v_3 are blue. To avoid a blue K_4 , v_3v_5 and v_2v_4 must now both be red. This forces v_2v_5 to be blue. Therefore, we can conclude, that up to relabeling of vertices and swapping of colours, the counterexample for $R(4, 4, C_5)$ is unique. Even when we select different edges to colour, we still obtain isomorphisms of this counterexample.

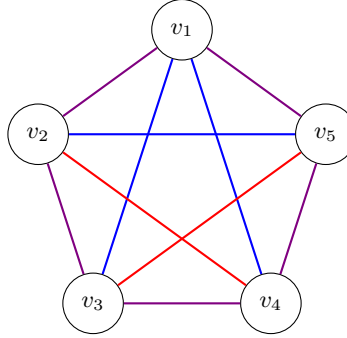


Figure 4: The K_5 counterexample with the colouring described in the proof

If we assume v_1v_4 and v_3v_5 to be blue, v_1v_3 must be red. Notably, regardless of the colours chosen for v_2v_4 and v_2v_5 , we find an isomorphism of the previous graph. If we make both edges red, we find the previous counterexample with swapped colours. Counterexamples where one edge is assumed to be red and one edge is assumed to be blue, leads to an isomorphism of the previous counterexample with relabelling of vertices. It must be noted that both edges cannot be the same colour, because this would lead to a monochromatic K_4 . This means that when adding vertices and exploring larger complete graphs with this constraint, such as K_6 and K_7 , we can assume that this K_5 is a subgraph without loss of generality.

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